

CURRICULUM VITAE JUAN AQUINO

Lima, Perú | juan.aquino@utec.edu.pe | +51 929 527 599
[linkedin.com/in/juan-aquino-espinoza](https://www.linkedin.com/in/juan-aquino-espinoza) | github.com/lavacacharles | **Universidad de Ingeniería y Tecnología**

EDUCATION

- Universidad de Ingeniería y Tecnología**

Lima, Perú | Mar 2022 - Present

B. S. Data Science — Top 10% of cohort. [Fully funded merit scholarship.](#)

 - Relevant Coursework: Analysis and Design of Algorithms, Optimization, Machine Learning, Generative Modeling, Geometric Deep Learning
 - Students Clubs: IISE (industrial engineering), Optiminds (competitive data science & ML)**Fundacao Getulio Vargas - Escola de Matemática Aplicada**

Rio de Janeiro, Brazil | Jan - Mar 2026

Programa do Verao at FGV-EMAP — [Fully funded international research scholarship.](#)

Relevant Coursework: Optimal transport, Algebraic structures, Group Theory

University of Melbourne

Melbourne, Australia | Incoming Jan 2027

Research Intern — [Research Experience for Peruvian Undergraduates.](#)

FELLOWSHIPS

- Databricks Student Fellowship Program**

2026

Selected as a Databricks Student Fellow among 36 (<1%) students chosen from over +5,800 applicants worldwide.

RESEARCH INTERESTS

Generative Modeling, Computer Vision, Scientific Data Visualization, Representational Learning, Data Science for Society.

RESEARCH EXPERIENCE

- Research Intern, Artificio**

Aug 2025 - Present

Advisor: MSc. Victor Flores, Artificio CTO

 - Implemented end to end Pytorch training pipelines for pretraining VQ GANs and Diffusion Transformers, parallelizing and reducing training time by 30%.
 - Automated data preprocessing and experiment tracking with OpenCV, reducing setup time per run by 2 hours and standardizing reproducibility for research workflows.
 - Experimented with flow group regularization and flow matching loss functions, improving generation quality and convergence stability by 10%
- Undergraduate Research Assistant - GINIA, Grupo de Investigación en Inteligencia Artificial**

Apr 2026 - Present

Advisor: PhD. Germain Zanabria, Research Associate Professor at UTEC

 - Designed and implemented end-to-end PyTorch training and inference pipelines, leveraging regularization techniques to address data scarcity.
 - Developed explainable transformer-based models by integrating causal mechanisms, enabling localization of perceptually unsafe regions in street-view images and enhancing insecurity score interpretability.
 - Built a segmentation pipeline using Panoptic and Cascade Mask R-CNN models to analyze urban scenes, improving insecurity score evaluation through correlations with key urban components.

PUBLICATIONS & ACTIVE RESEARCH PROJECTS

- Gas and Electricity Production Forecasting**

2024

Published in the Journal Ginia, University AI Research Group
Supervised by Cristian Lopez del Álamo, Research Associate Professor at UTEC

 - Conducted exploratory analysis of energy-related variables to identify dependencies with industrial gas and electricity production using pandas, matplotlib, and tslearn.
 - Developed a custom NumPy implementation of forecasting models based on the optimization of state-space models (SSMs) and SARIMA, evaluated using AIC and BIC criteria.
 - Assessed model robustness and interpretability through statistical testing and residual analysis, characterizing the model's ability to capture underlying dynamic patterns.
- Graph Expressiveness Loss via homomorphisms encoding**

2026

Advisor: MSc. Ariana Villegas, Associate Professor at UTEC

 - Characterized graph datasets on node features and graph structure to identify complementary task relevant information, and to enhance clear explainable properties of graph datasets.

- Trained graph neural networks based on convolutions (GCN, GAT, GIN) and homomorphism counting methods to compare layer by layer expressiveness loss identifying graph representation properties.
- Provide insights through homomorphisms distortion metric about model performance without training on downstream tasks by exploring sub graph structure patterns, clarifying extracted hidden patterns by the models.

World Models for Driving Behavior Transfer

2025

Advisor: MSc. Victor Flores & PhD. Arturo Deza

- Developed extensive tests of vector quantized generative adversarial networks, handling instability dynamics and quality degradation to emulate urban scenes.
- Experimented with diffusion regularization and quality recovery methods to reconstruct and enhance robustness on long horizon generation for ego driving videos.
- Characterized traffic chaotic scenes using information retrieval on peruvian chaos driving dynamics and tested world models on generalization of real physics simulation.

Autonomous Driving Model in Carla Environment

2025

First place in the Deep Learning 2025-I course competition

- Developed an end to end driving pipeline within the Carla simulator by computer vision models.
- Implemented augmentation techniques to expand the field of view of autonomous driving models using attention mechanisms.
- Trained and validated autonomous driving models reaction capabilities using challenging driving scenarios, such as intersections with traffic lights, pedestrians, turns, etc.

INDUSTRY EXPERIENCE

Machine Learning Teaching Assistant, UTEC

Apr 2026 - Present

Supervised by Victor Martinez, Associate Professor at UTEC

- Elaborated exams and course notes to support students practice
- Attended to laboratories assessing students and clarifying course concepts

Optimization Students Mentor, UTEC

Apr 2026 - Present

Supervised by Victor Torres Castillo, Associate Professor at UTEC

- Elaborated exams and course notes to support students practice
- Attended to laboratories assessing students and clarifying course concepts

Strategy Engagement Intern, DIDI Foods

Oct 2025 - Jan 2026

Supervised by Renato Carbajal Alván, Sr Strategy Analyst

- Analyzed historical campaigns in Excel through quantitative diagnosis for ROI optimization of campaign strategies, leveraging a 13% of savings.
- Evaluated user demand features for the segmentation of geographic locations in three countries to identify key investment opportunities.

LEADERSHIP

Project Manager & Hackathon Trainer - Optiminds

- Designed and delivered training programs for 30+ students covering end-to-end ML pipelines, agentic automation systems, and technical communication.
- Led 10 interdisciplinary teams to compete in national hackathons (BECAS BCP, DATATÓN, ENERGÍA, DATAFEST), advancing Peruvian student participation in competitive data science.

SKILLS

- **Languages** English: fluent, Spanish: Native language
- **Coding Languages** Python, C++, javascript, SQL, R
- **Frameworks/Libraries** Pytorch, Polars, Numpy, Cupy, Dask, Pytorch lightning, Tensorflow, Docker, Slurm, FastAPI
- **Tools/Technologies** AWS, Google Cloud Platform, Snowflake, MongoDB, Neo4j, Git, Linux, Excel